Darasimi Ajayi

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Education

Dalhousie University | GPA: 4.17/4.30 — Halifax, NS
Bachelor of Computer Science — *Jan 2023 – Jan 2027*

Experience

Software Developer — May 2025 – Aug 2025
Nautel Ltd — *Halifax, NS*

- Enhanced automated testing infrastructure within an **Agile** team, reducing QA workload and improving test coverage by 25%.
- Authored and maintained 150+ end-to-end tests using **Typescript**, **Playwright** and **Cucumber**, expanding coverage and preventing regressions.
- Created and standardized 200+ unique Test IDs across application pages to streamline test creation and reduce QA/developer ambiguity.
- Implemented automated Git hooks (via **Husky**) to enforce **code quality** checks on every commit and push.

Full Stack Developer — September 2024 – Dec 2024
SimplyCast — *Halifax, NS*

- Individually resolved 50+ bug tickets, contributing to backlog reduction from 200+ issues to 90 as part of a 3-developer **Agile** team.
- Diagnosed and fixed complex **PHP** and **JavaScript** issues using **Chrome DevTools** and **Xdebug**, reducing debugging time by 30%.
- Resolved issues in **Angular**, **Vue**, and **PHP** components, modifying **MySQL** queries to fix bugs and enhance system stability.
- Collaborated with senior developers on complex issues, seeking feedback during solution planning to strengthen **code quality**.
- Developed, modified, and integrated **API** endpoints to resolve software bugs and ensure smoother data flow.

Projects

DaraOS (DOS) | ***Rust, QEMU***, *GDB* — Jan 2026 – Present

- Developing a custom cross-platform operating system kernel from scratch in **Rust**, targeting both **x86_64** workstations and **ARM** microcontrollers.
- Implemented a **no_std** environment, managing low-level hardware abstractions, memory-mapped I/O, and custom bootloader configurations.
- Built a **VGA text buffer driver** to handle screen output and integrated a **serial port interface** for kernel debugging and logging.
- Leveraged **QEMU** for hardware emulation and **GDB** for kernel-level debugging, ensuring stability across different CPU architectures.

3D Marine Environment Simulation | ***C++, OpenGL, GLSL***, *Linear Algebra* — Nov 2024 – Dec 2024

- Developed a 3D submarine simulation utilizing **OpenGL** for the final course project, implementing **Phong lighting models** and material properties to achieve realistic visual effects.
- Engineered a dynamic water wave simulation using **sine-based vertex displacement**, calculating real-time surface normals for accurate light reflection and refraction.
- Integrated an external **model loading** pipeline to render complex submarine geometries with **UV texture mapping** and coordinate space transformations.
- Optimized rendering performance by managing **Vertex Buffer Objects (VBOs)** and **Shaders**, ensuring smooth frame rates during real-time movement simulation.

16-Bit CPU | *CircuitVerse*, ***C*** — Feb 2023 – March 2023

- Designed and built a simulated 16-bit CPU from scratch using **CircuitVerse**, implementing 15 instructions with an extensible **architecture**.
- Implemented instruction logic manually with **Boolean algebra** and digital logic gates, constructing components such as the **ALU**, registers, and control unit.
- Validated all components and instructions through test cases, ensuring correctness and reliability of execution.
- Developed a custom **assembly language** in **C** to program the CPU, enabling demonstration programs and low-level applications.

Technical Skills

Languages: C, C++, Java, Python, C#, PHP, JavaScript, Typescript, Rust, MySQL
Frameworks & Libraries :SDL3, OpenGL4.6, Spring Boot, Vue.js, AngularJS, Playwright, Cucumber